

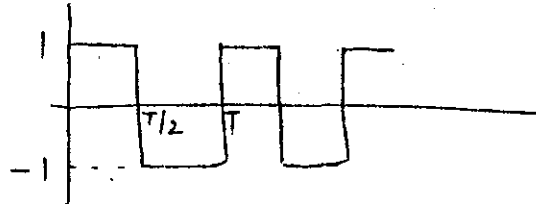
B.Tech Degree VI Semester Examination May 2003

ME 605 CONTROL SYSTEM ENGINEERING (1999 Admissions onwards)

Time: 3 Hours

Maximum Marks: 100

- I. (a) Find the Laplace Transform of the waveform shown below :



(10)

- (b) Solve the following differential equation by Laplace Transform method.

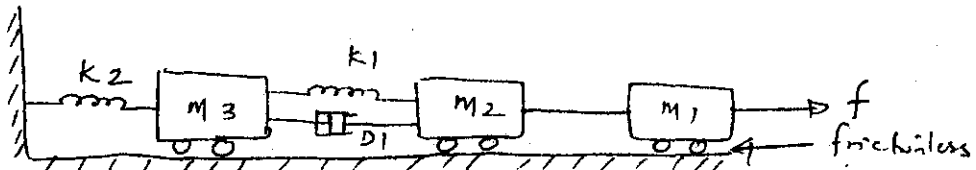
$$\frac{d^2x}{dt^2} + 2\frac{dx}{dt} + 5x = 3 \text{ given that } x(0) \text{ and } \dot{x}(0) = 0.$$

(10)

OR

- II. (a) Draw a typical step response of a second order system and show the time domain specifications. (5)
- (b) What is meant by transient and steady state response ? What are the 3 static error constants ? (5)
- (c) Consider the unity feedback systems with forward transfer function given by $G(s) = \frac{25}{s(s+5)}$. Obtain rise time t_r , peak time t_p , maximum overshoot M_p and settling time t_s when it is subjected to a unit step input. (10)

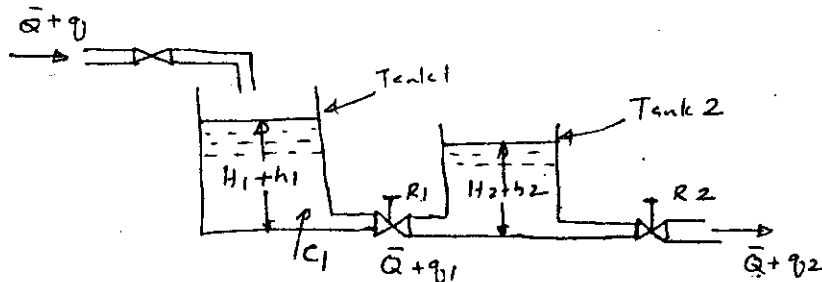
- III. (a) Explain force-voltage and force-current analogy. (8)
- (b) Draw the electrical analog of the mechanical system given below:



(12)

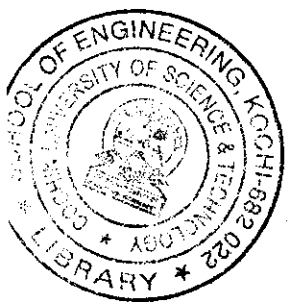
OR

- IV. (a) What is meant by state of a system ? Explain what are state variables and state model of a system. (10)
- (b) Determine the transfer function of the liquid level system shown in figure.



(10)

(Turn Over)



- V. (a) What is meant by stability ? What are the necessary conditions for a system to be stable ? (6)
 (b) Explain Routh Hurwitz criterion for determining the stability of a system. (6)
 (c) Check whether the system is stable or not.
 $s^4 + 8s^3 + 18s^2 + 16s + 5 = 0$ (8)
- OR**
- VI. (a) What is meant by Frequency response ? Explain the frequency response characteristics. (8)
 (b) Draw the Bode plot of the system given by transfer function
 $G(s) = \frac{64(s+2)}{s(s+0.5)(s^2+3.2s+64)}$ and $H(s) = 1$. (12)
- VII. (a) What is a discrete signal ? Explain sampling. What is sampling theorem ? (8)
 (b) Determine the impulse response of the Linear Discrete system described by
 $c(k) - \alpha c(k-1) = r(k)$ (6)
 (c) Find the Z- transform of the output of the Linear Discrete System given by
 $c(k+2) + a_1 c(k+1) + a_2 c(k) = b_0 r(k+2) + b_1 r(k+1) + b_2 r(k)$ (6)
- OR**
- VIII. (a) Explain Z - and S - domain Relationship. (6)
 (b) Explain how will you determine stability by Jury's Test. (7)
 (c) Determine the stability of the system given by
 $F(Z) = 2 Z^4 + 7 Z^3 + 10 Z^2 + 4 Z + 1$ (7)
- IX. (a) What is the need for a pneumatic equipment in industry ? Draw a pneumatic flapper amplifier. (10)
 (b) What is Integral and Derivative Control ? Explain the effects of these controllers in the system performance. (10)
- OR**
- X. Write short notes on the following :
 (i) synchros
 (ii) stepper motors
 (iii) tachogenerators
 (iv) DC and AC servomotors. (20)
